MH2500 Probability and Introduction to Statistics

Slides for Week 9-1

Topics: More on Joint Distributions

Wu Guohua

Nanyang Technological University

Example

The joint density of X and Y is given by

$$f(x,y) = \begin{cases} e^{-(x+y)}, & 0 < x, y < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Find the density function of the random variable $Z = X/Y$.

Solution: We find the CDF first and then differentiate it to get the pdf. For $z > 0$,

$$P\{Z \leq z\} = P\{X/Y \leq z\} = P\left\{Y \geq \frac{X}{z}\right\} = \int_0^\infty \int_{x/z}^\infty e^{-x} e^{-y} dy dx$$

$$= \int_0^\infty \left[e^{-x} \left(-e^{-y}\right)_{x/z}^\infty\right] dx = \int_0^\infty e^{-x} e^{-x/z} dx$$

$$= \int_0^\infty e^{-x(1+1/z)} dx = \left(-\frac{1}{1+1/z} e^{-x(1+1/z)}\right)_0^\infty = \frac{1}{1+1/z} = \frac{z}{1+z}.$$

We find the CDF first and then differentiate it to get the pdf.

- For $z \leq 0$, $f_Z(z) = 0$.
- For $z > 0$,

$$P\{Z \leq z\} = P\left\{Y \geq \frac{X}{z}\right\} = \frac{z}{1+z}.$$ Differentiate it, we have:

$$f_Z(z) = \frac{d}{dz}P\{Z < z\} = \frac{d}{dz}\left(1 - \frac{1}{1+z}\right) = \frac{1}{(1+z)^2}.$$

Independent Random Variables

Say that two random variables X and Y are independent $\underline{\underline{\text{if and only if}}}$

- $P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}$, for any "reasonable" A and B;
- $F(x, y) = F_X(x)F_Y(y)$ for any $x, y \in \mathbb{R}$.

- discrete:

$$p(x, y) = p_X(x)p_Y(y);$$

- continuous:

$$f(x, y) = f_X(x)f_Y(y).$$

Example

Suppose that a node in a communication network has the property that if two packets of information arrive within time τ of each other, they 'collide' and then have to be retransmitted.

If the times of arrival of the two packets are independent and uniform on $[0, T]$,

- what is the probability that they collide?

(collision means that $|T_1 - T_2| \leq \tau$)

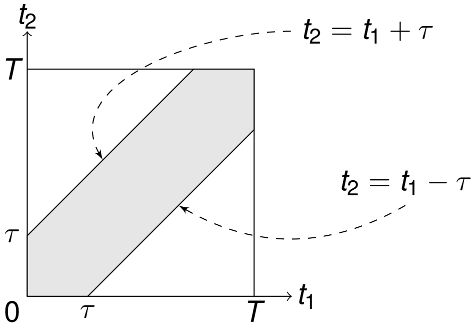
The times of arrival of two packets, T_1 and T_2 are independent and uniform on $[0, T]$, so their joint density is the product of the marginals, or

$$f(t_1, t_2) = \frac{1}{T^2}$$

for t_1 and t_2 in the square with sides $[0, T]$.

Therefore, (T_1, T_2) is uniformly distributed over the square.

The probability that the two packets collide is proportional to the area



Area of shaded area = $T^2 - (T - \tau)^2$. Hence required probability is

$$\frac{T^2 - (T - \tau)^2}{T^2} = 1 - \left(1 - \frac{\tau}{T}\right)^2.$$

Proposition

X and Y are independent if and only if their joint probability density/mass function can be expressed as

$$f_{X,Y}(x,y) = h(x)g(y), \quad -\infty < x, y < \infty.$$

Example

Determine whether X and Y are independent if their joint density is

- $f(x,y) = \begin{cases} 6e^{-2x}e^{-3y} & \text{when } x > 0, y > 0, \\ 0 & \text{otherwise;} \end{cases}$
- $g(x,y) = \begin{cases} 24xy & \text{when } 0 < x < 1, 0 < y < 1, 0 < x + y < 1, \\ 0 & \text{otherwise;} \end{cases}$

Proposition

Suppose X and Y are independent. If g and h are functions, then $U = g(X)$ and $V = h(Y)$ are independent as well.

Sketch of argument:

Let $A(u) = \{x : g(x) \leq u\}$ and $B(v) = \{y : h(y) \leq v\}$. Then

$$\begin{aligned} P\{U \leq u, V \leq v\} &= P\{X \in A(u), Y \in B(v)\} \\ &= P\{X \in A(u)\}P\{Y \in B(v)\} \\ &= P\{U \leq u\}P\{V \leq v\}. \end{aligned}$$

Sum of Continuous Random Variables

Recall that when X and Y are independent and discrete, Z is discrete and has the probability mass function

$$p_Z(z) = \sum_x p_X(x) p_Y(z - x).$$

Similarly,

Proposition

When X and Y are independent continuous random variables, $Z = X + Y$ has the probability density function

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(x) p_Y(z - x) dx.$$

We first find the cdf of $Z = X + Y$ and differentiate to find the density. Since $Z \leq z$ whenever the point (X, Y) is in the region $R_z = \{(x, y) : x + y \leq z\}$, we have

$$F_Z(z) = \iint_{R_z} f(x, y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f(x, y) dy dx.$$

Making the change of variables $y = v - x$ in the inner integral gives

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z} f(x, v - x) dv dx = \int_{-\infty}^{z} \int_{-\infty}^{\infty} f(x, v - x) dx dv.$$

Differentiate it, and if $\int_{-\infty}^{\infty} f(x, z-x)dx$ is continuous at z, then

$$f_Z(z) = \int_{-\infty}^{\infty} f(x, z-x)dx.$$

- If X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)dx.$$

- This integral is called the **convolution** of the functions f_X and f_Y.

Sum of Uniform Random Variables

Example

Let X and Y be independent uniform random variables on $(0, 1)$. Find the density of $Z = X + Y$.

Solution: The density of a $Unif(0, 1)$ variable is $f(z) = 1$ on $[0, 1]$. The density function of Z is

$$f_Z(z) = \int_{-\infty}^{\infty} f(x)f(z-x)dx = \int_0^1 f(z-x)dx$$

The integrand is nonzero only when $0 \leq z - x \leq 1$, i.e.,

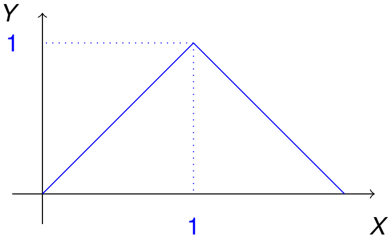
$$z - 1 \leq x \leq z.$$

- $z \leq 1$,

$$f_Z(z) = \int_0^z dx = z,$$

- $1 \leq z \leq 2$,

$$f_Z(z) = \int_{z-1}^1 dx = 2 - z.$$



Gamma Distribution Revisited

In the model of phone calls to a service centre

- it is assumed that the waiting time between two consecutive calls follows $\mathsf{Exp}(\lambda)$
- we derived that the waiting time until the nth call follows $\mathsf{Gamma}(n, \lambda)$

Now let us verify this directly, without using Poisson distribution.

Proposition

Suppose that $X_1, \ldots, X_n$ are i.i.d. $\mathsf{Exp}(\lambda)$ random variables. Then $X_1 + \cdots + X_n \sim \mathsf{Gamma}(n, \lambda)$.

Sum of Gamma Variables

Since $\mathsf{Exp}(\lambda)$ is $\mathsf{Gamma}(1, \lambda)$, it suffices to prove the following:

Proposition

Suppose that $X \sim \mathsf{Gamma}(\alpha, \lambda)$ and $Y \sim \mathsf{Gamma}(\beta, \lambda)$ are independent. Then $X + Y \sim \mathsf{Gamma}(\alpha + \beta, \lambda)$.

Recall the density of $\mathsf{Gamma}(t,\lambda)$: $p(x) = C_{t,\lambda} e^{-\lambda x} x^{t-1}$, $x \geq 0$.

$$
\begin{aligned}
f_{X+Y}(z) &= \int_{-\infty}^{\infty} p_X(x) p_Y(z-x) dx \\
&= \int_0^z C_{\alpha,\lambda} e^{-\lambda x} x^{\alpha-1} \cdot C_{\beta,\lambda} e^{-\lambda(z-x)} (z-x)^{\beta-1} dx \\
&= Ce^{-\lambda z} \int_0^z x^{\alpha-1} (z-x)^{\beta-1} dx \\
&= Ce^{-\lambda z} \int_0^1 (uz)^{\alpha-1} z^{\beta-1} (1-u)^{\beta-1} \ zdu \quad (\text{let } u = x/z) \\
&= Ce^{-\lambda z} z^{\alpha+\beta-1} \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du \\
&= C' e^{-\lambda z} z^{\alpha+\beta-1}
\end{aligned}
$$

Therefore, $X + Y \sim \mathsf{Gamma}(\alpha+\beta, \lambda)$.